INCH-POUND
MIL-DTL-24643/35E
27 November 2023
SUPERSEDING
MIL-DTL-24643/35D
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LS3SU (INCLUDING VARIATION LS3SUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS3SU and LS3SUS (double overall shield).

Construction (Non-Watertight)

First	-	Copper conductor, coated or uncoated, 18 AWG (7 by 0.0152 inches).
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Nominal wall thickness of 0.013 inch; wall thickness may vary from nominal as necessary in order for the completed cable to meet the specified electrical requirements. Colored insulation, one white, one black, and one red conductor to form each triad.
Fourth	-	Three conductors (one black, one white, and one red) cabled together with a nominal lay of 3.5 inches to form each triad.
Fifth	-	An optional binder.
Sixth	-	Braided shield of 34 or 36 AWG, coated or uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Seventh	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape, plus an extruded jacket of a suitable transparent material; minimum average thickness of the transparent material 0.003 inch. Standard identification code shall be applied by method 2.
Eighth	-	The specified number of triads (see table I) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.
Ninth	-	An optional binder.

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Construction for Variant Type LS3SU

Tenth - Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Construction for Variant Type LS3SUS

Tenth - Braided shield of tin-coated copper.

Eleventh - Spirally wrapped, 0.002-inch-thick polyethylene terephthalate (polyester film) separator tape with a 25-percent minimum overlap.

Twelfth - Braided shield of tin-coated copper.

Thirteenth - An optional binder.

Fourteenth - Cross-linked polyolefin jacket. See table II for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details (unshielded).

Military part no. M24643/35	Type and size	Number of triads	Cable jacket thickness min. avg. (inch)	Overall diameter (inches)	
				min.	max.
-01UN	LS3SU-3	3	0.040	0.650	0.700
-02UN	LS3SU-7	7	0.050	0.840	0.910
-03UN	LS3SU-10	10	0.050	1.100	1.190
-04UN	LS3SU-14	14	0.050	1.200	1.290
-05UN	LS3SU-19	19	0.050	1.340	1.430
-06UN	LS3SU-24	24	0.060	1.580	1.670
-07UN	LS3SU-30	30	0.060	1.680	1.770
-08UN	LS3SU-37	37	0.060	1.840	1.930
-09UN	LS3SU-44	44	0.060	2.060	2.150

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TABLE II. Details (shielded).

Military part no. M24643/35	Type and size	Number of triads	Cable jacket thickness min. avg. (inch)	Overall diameter (inches)	
				min.	max.
-01UD	LS3SUS-3	3	0.040	0.710	0.760
-02UD	LS3SUS-7	7	0.050	0.900	0.970
-03UD	LS3SUS-10	10	0.050	1.160	1.250
-04UD	LS3SUS-14	14	0.050	1.260	1.350
-05UD	LS3SUS-19	19	0.050	1.400	1.490
-06UD	LS3SUS-24	24	0.060	1.640	1.730
-07UD	LS3SUS-30	30	0.060	1.740	1.830
-08UD	LS3SUS-37	37	0.060	1.900	1.990
-09UD	LS3SUS-44	44	0.060	2.120	2.210

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	7.06
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Conductor to shield	1,000
Component shield to all other shields	500
Overall shield to overall shield (type LS3SUS only)	200
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Component shield to all other shields	100
Overall shield to overall shield (type LS3SUS only)	100
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws (type LS3SUS only)	No failure

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Group A:	
Visual and dimensional	No failure
Capacitance	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 feet)	30
Capacitance unbalance (percent, max.)	10
Group B:	
Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Shield (conformance to material, construction, and coverage)	No failure
Surface transfer impedance (type LS3SUS only)	

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Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

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UNIT ORDERING LENGTHS:

Types	Sizes	Feet (nominal)
LS3SU and LS3SUS	3 through 19	1,000
LS3SU and LS3SUS	24 through 44	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-051)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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